

SOLUTION

ASSESSMENT TOOL 1 – CASE STUDY SOLUTION

Course Name: Artificial Intelligence

Course Code: CSA1717

Course Outcome Covered: CO3 – Knowledge Representation and Reasoning

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1. CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

1.1 Problem Statement

A hospital uses a rule-based logical agent to identify possible diseases from patient symptoms. The system applies Propositional Logic, Forward Chaining, and Backward Chaining.

Patient A:

- Fever = True
- Cough = True
- Breathlessness = True

1.2 Knowledge Representation

Let:

- F = Fever
- C = Cough
- R = Rash
- B = Breathlessness
- Flu = Possible Flu
- Measles = Possible Measles
- P = Possible Pneumonia

Rules

R1: $F \wedge C \rightarrow \text{Flu}$

R2: $F \wedge R \rightarrow \text{Measles}$

R3: $C \wedge B \rightarrow P$

Facts

F = True, C = True, B = True

1.3 Forward Chaining

Step	Rule	Conditions	Result
0		Initial F, C, B	Known facts
1	R1	$F \wedge C$	Flu
2	R2	$F \wedge R$	Cannot fire
3	R3	$C \wedge B$	Pneumonia

Final Result

Flu = True

Pneumonia = True

Measles = Not Concluded

1.4 Backward Chaining

Goal: Pneumonia

Rule:

$C \wedge B \rightarrow \text{Pneumonia}$

Sub-goals:

Cough $\rightarrow$ True

Breathlessness $\rightarrow$ True

Both conditions are available as facts.

Therefore:

Pneumonia is successfully proved.

1.5 Conclusion

Forward Chaining derives Flu and Pneumonia from the available symptoms. Backward Chaining verifies Pneumonia because Cough and Breathlessness are known facts.

2. CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

2.1 Knowledge Base

R1: $\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

R2: $\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

R3: $\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

Facts

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

$\text{Vehicle}(\text{CarA})$

$\text{Behind}(\text{CarA}, \text{Ambulance})$

2.2 Unification

Match:

Vehicle(x) with **Vehicle(Ambulance)**

Therefore:

$\theta = \{x / \text{Ambulance}\}$

After substitution:

$\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance})$
 $\rightarrow \text{ClearPath}(\text{Ambulance})$

Hence:

ClearPath(Ambulance)

Using R2:

GreenSignal(Ambulance)

Using R3:

$\text{Vehicle}(\text{CarA}) \wedge \text{Behind}(\text{CarA}, \text{Ambulance}) \wedge \text{GreenSignal}(\text{Ambulance})$
 $\rightarrow \text{Proceed}(\text{CarA})$

Therefore:

Proceed(CarA) = True

2.3 Resolution

CNF Conversion

R1:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

R2:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

R3:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Resolution Proof

Negate the goal:

$\neg \text{Proceed}(\text{CarA})$

From the facts and rules:

Vehicle(Ambulance)

- EmergencyType(Ambulance)
→ ClearPath(Ambulance)

ClearPath(Ambulance)

→ GreenSignal(Ambulance)

Vehicle(CarA)

- Behind(CarA, Ambulance)
- GreenSignal(Ambulance)
→ Proceed(CarA)

Therefore:

Proceed(CarA)

- $\neg \text{Proceed}(\text{CarA})$
→ □

The empty clause proves:

Proceed(CarA) is TRUE.

2.4 Multiple Emergency Vehicles

The current KB can process multiple emergency vehicles if their required facts are available. However, additional rules are needed for:

- Emergency priority
- Route conflicts
- Signal scheduling
- Vehicle location
- Direction and destination

Example:

$\text{EmergencyVehicle}(x) \wedge \text{HigherPriority}(x,y) \rightarrow \text{GivePriority}(x)$

2.5 Conclusion

FOL, Unification, and Resolution successfully prove that CarA can proceed. Additional priority and routing rules are required for simultaneous emergencies.

3. CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

3.1 Knowledge Base

Let:

- $S = \text{SoilDry}$
- $I = \text{IrrigationNeeded}$
- $W = \text{CropWheat}$
- $D = \text{ApplyDripMethod}$
- $R = \text{CropAtRisk}$

Rules

P1: $S \rightarrow I$

P2: $I \wedge W \rightarrow D$

P3: $\neg D \wedge W \rightarrow R$

Facts

S, W

3.2 CNF Conversion

P1

$S \rightarrow I$

$\neg S \vee I$

P2

$I \wedge W \rightarrow D$

$\neg I \vee \neg W \vee D$

P3

$\neg D \wedge W \rightarrow R$

$D \vee \neg W \vee R$

Final CNF

- $C1: \neg S \vee I$

- C2: $\neg I \vee \neg W \vee D$
- C3: $D \vee \neg W \vee R$
- C4: S
- C5: W

3.3 Resolution Refutation

Goal: D

Negated goal:

$\neg D$

Step 1

C1 + C4:

I

Step 2

C2 + I:

$\neg W \vee D$

Step 3

C5 + ($\neg W \vee D$):

D

Step 4

D + $\neg D$:

□

Therefore:

ApplyDripMethod is proved.

3.4 Removing SoilDry

If **SoilDry** is removed:

- IrrigationNeeded cannot be derived.
- ApplyDripMethod cannot be derived.
- CropAtRisk cannot be concluded.

The absence of ApplyDripMethod does not mean $\neg$ ApplyDripMethod.

3.5 Conclusion

Resolution proves that ApplyDripMethod follows when SoilDry and CropWheat are known. Removing SoilDry breaks the inference chain.

4. CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

4.1 Given Percepts

- Stench[1,2]
- Breeze[1,1]
- Glitter[2,2]

4.2 Modus Ponens

Step 1

Stench[1,2]

→ Wumpus is adjacent to [1,2]

Step 2

Breeze[1,1]

→ Pit is adjacent to [1,1]

Step 3

Glitter[2,2]

→ Gold[2,2]

4.3 Belief State

The agent knows:

- Stench[1,2]
- Breeze[1,1]
- Glitter[2,2]
- Gold[2,2]
- Wumpus is adjacent to [1,2]
- Pit is adjacent to [1,1]

4.4 Forward Chaining

Possible Wumpus Locations

Adjacent cells to [1,2]:

[1,1], [1,3], [2,2]

Therefore, possible Wumpus locations are:

[1,1], [1,3], [2,2]

Possible Pit Locations

Adjacent cells to [1,1]:

[1,2], [2,1]

Therefore, possible Pit locations are:

[1,2], [2,1]

The exact locations cannot be determined.

4.5 Path Safety

Path:

[1,1] → [1,2] → [2,2]

Glitter[2,2] proves:

Gold[2,2]

However, there are no explicit facts proving:

¬Wumpus[2,2]

and

¬Pit[2,2]

Therefore, **Safe[2,2] cannot be proved.**

Final Decision

The path **cannot be guaranteed to be safe** using the given knowledge base.

4.6 Conclusion

The agent successfully identifies the gold location and possible hazard locations. However, incomplete information prevents the system from guaranteeing that the complete path is safe.

6. REFERENCES

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